CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY & RESEARCH

Department of Information Technology

Subject Name: Data Engineering

Semester: 5th

Subject Code: ITUE301

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Practical File

PRACTICAL-1

Problem Definition:

Study, dissect, and map the complete end-to-end Data Engineering Lifecycle for an enterprise e-commerce platform. Students must trace data states across all five core lifecycle stages: Data Generation, Data Ingestion, Data Storage, Data Transformation, and Data Serving. The solution requires creating a detailed architectural data flow diagram along with a system manifesto documenting cross-cutting lifecycle undercurrents, including Security, Observability, and Data Privacy. The completed structure should be committed to a personal Git repository.

Key Questions:

● **Que / Key Point 1:** How do the technical demands of downstream serving layers directly influence ingestion and storage design decisions during the early generation phases?

1. Determine whether data should be ingested in real time (streaming) or in batches based on application requirements.
2. Select an appropriate data format such as structured, semi-structured, or unstructured for efficient processing.
3. Choose a suitable storage solution, including transactional databases, data lakes, or data warehouses.
4. Apply data validation and quality checks during ingestion to ensure accurate downstream processing.
5. Improve serving performance using indexing, partitioning, and optimized query strategies.
6. Design the ingestion and storage layers to support scalability, reliability, and future business growth.

● **Que / Key Point 2:** Differentiate between the broader data lifecycle and the technical data engineering lifecycle within an enterprise.

Data Lifecycle	Data Engineering Lifecycle
Covers the complete journey of data.	Focuses on technical processing of data.
Business-oriented.	Technology-oriented.
Includes creation, usage, archival, and deletion.	Includes ingestion, storage, transformation, and serving.
Involves business and compliance teams.	Involves data engineers and architects.

● **Que / Key Point 3:** Identify how the undercurrents of security and data observability should be enforced at each lifecycle boundary.

1. Data Generation: Secure data collection using authenticated sources and monitor incoming data quality.

2. Data Ingestion: Encrypt data during transmission, validate incoming records, and monitor pipeline performance.

3. Data Storage: Protect stored data using encryption, role-based access control, backup policies, and continuous health monitoring.

4. Data Transformation: Ensure secure data processing, validate transformed data, and monitor processing accuracy.

5. Data Serving: Authenticate users, authorize data access, monitor query performance, and maintain audit logs for data access.

Supplementary Problems:

Map out the data engineering lifecycle for a smart city IoT traffic sensor platform, emphasizing challenges related to continuous real-time streaming data generation.

1. Data Generation

- Traffic sensors continuously generate vehicle and road data.
- GPS, cameras, and IoT devices produce real-time streams.
- **Challenge:** High data volume and continuous data generation.

2. Data Ingestion

- Streaming platforms collect sensor data in real time.
- Handle thousands of events per second.
- **Challenge:** Low latency and reliable data transmission.

3. Data Storage

- Store raw data in a data lake and processed data in databases.

- Partition data based on location and time.
- **Challenge:** Scalable storage for massive datasets.

4. Data Processing / Transformation

- Clean, validate, and aggregate sensor data.
- Remove duplicate or missing records.
- **Challenge:** Real-time processing with minimal delay.

5. Data Serving

- Deliver live traffic updates to dashboards and navigation apps.
- Provide APIs for traffic management systems.
- **Challenge:** Fast query response and high availability.

6. Data Consumption

- Used by traffic authorities, commuters, and emergency services.
- Supports congestion analysis and route optimization.
- **Challenge:** Accurate and up-to-date information.

7. Monitoring & Maintenance

- Monitor sensor health and pipeline performance.
- Detect failures and data anomalies.
- **Challenge:** Ensuring continuous operation and data quality.

Major Challenges of Continuous Real-Time Streaming

• Managing high-speed continuous data generation.

• Ensuring low-latency data ingestion and processing.

• Handling network failures and preventing data loss.

• Providing scalable storage and processing infrastructure.

• Maintaining data security, privacy, and regulatory compliance.

• Continuously monitoring system health and data quality.

OUTPUT:

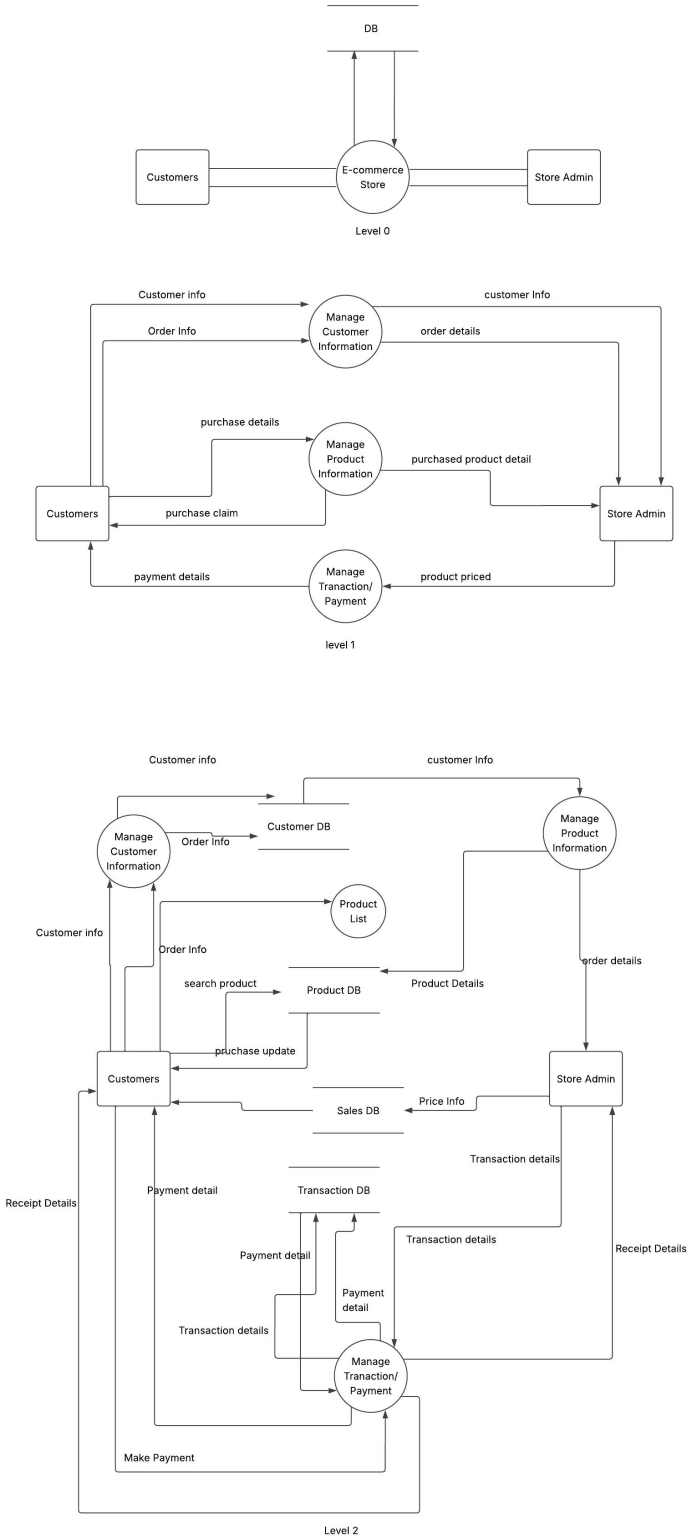


Figure 1: Enterprise E-Commerce Data Flow Diagram (Level 0, Level 1, and Level 2)

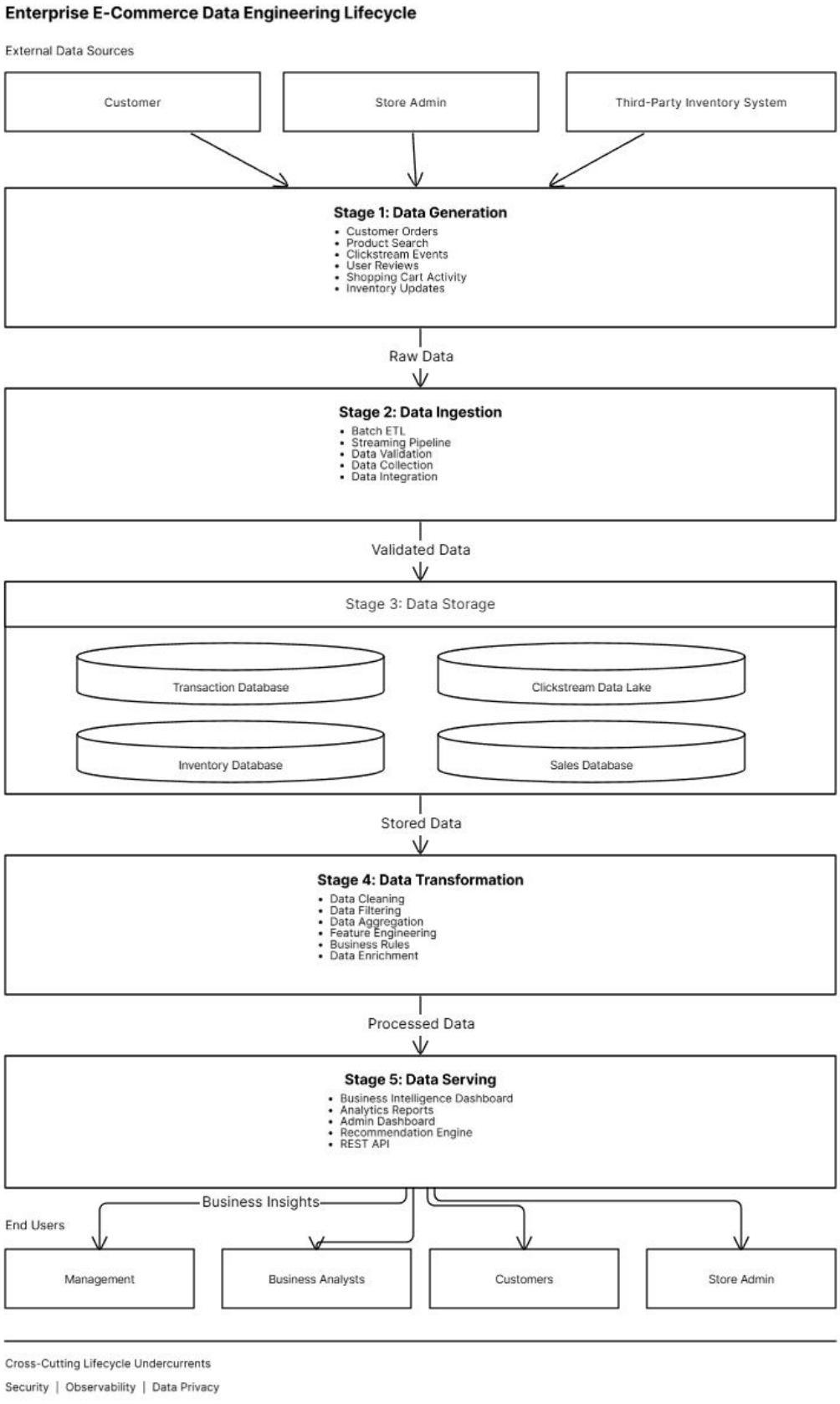


Figure 2: Enterprise E-Commerce Data Engineering Lifecycle Diagram